West Windbreak as One Guild Row — Mock-up

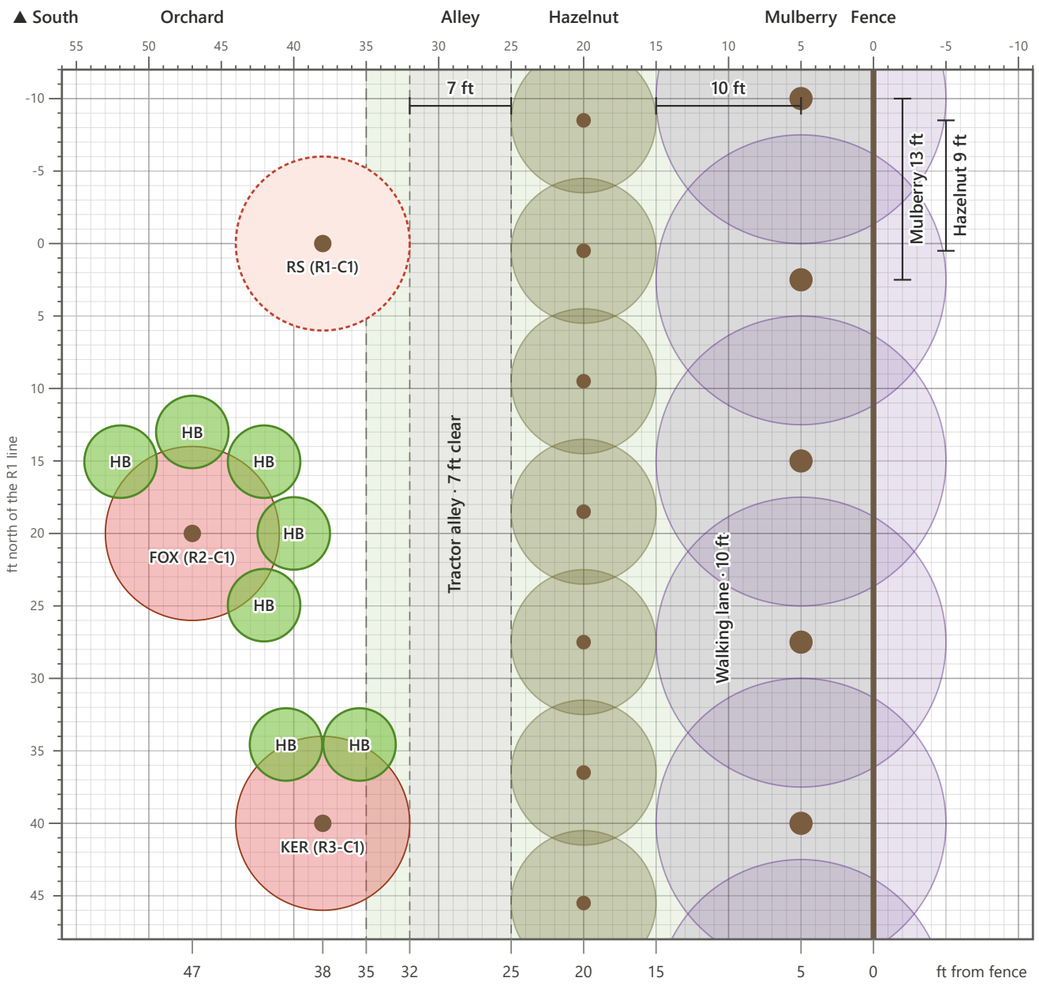
One row at the west fence instead of two: red mulberries with hazelnuts between them, and a row of small berries like Josta in front. The 10 ft walking lane and the separate hazelnut row go away, and the tractor alley gets the room.

Mock-up only. The planting plan, the data files, and every other page are unchanged. The current two-row windbreak is on the orchard pollination grid; the left side of each pair of figures below is drawn from that same data, at the same scale. To see all three patterns side by side, and how much sun the hazelnuts get, open the options and light check.

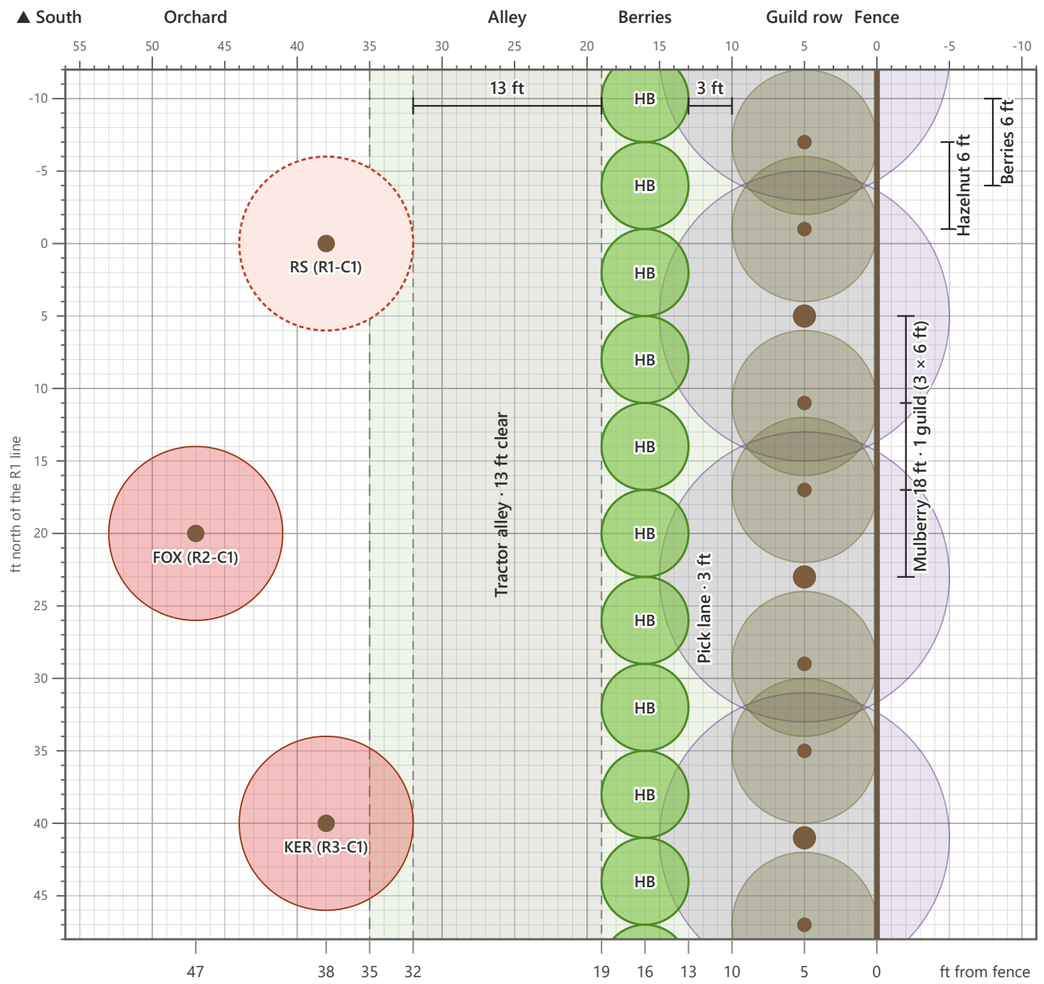
STRIP USED	TRACTOR ALLEY	GROUND FREED	RED MULBERRY	HAZELNUT	SMALL BERRIES IN THE STRIP	SUN ON THE HAZELNUTS
25 → 19 ft	**7 → 13 ft clear**	**about 1,750 sq ft**	**25 → 18**	**34 → 34**	**0 → 41**	**74% → 70%**
from the fence to the last row of plants	6 → 12 ft if every plant reaches its full width	6 ft along 292 ft of fence, all of it added to the alley	7 of the 25 on hand spare	all 34 on hand used	the bushes now planned around the C1 trees; 10 more places open	55% → 38% if the mulberries are left to spread

Plan view — to scale, south up, west fence on the right

Now: two rows and a walking lane (the current plan)



Proposed: pattern B, two hazelnuts between mulberries (M–H–H)

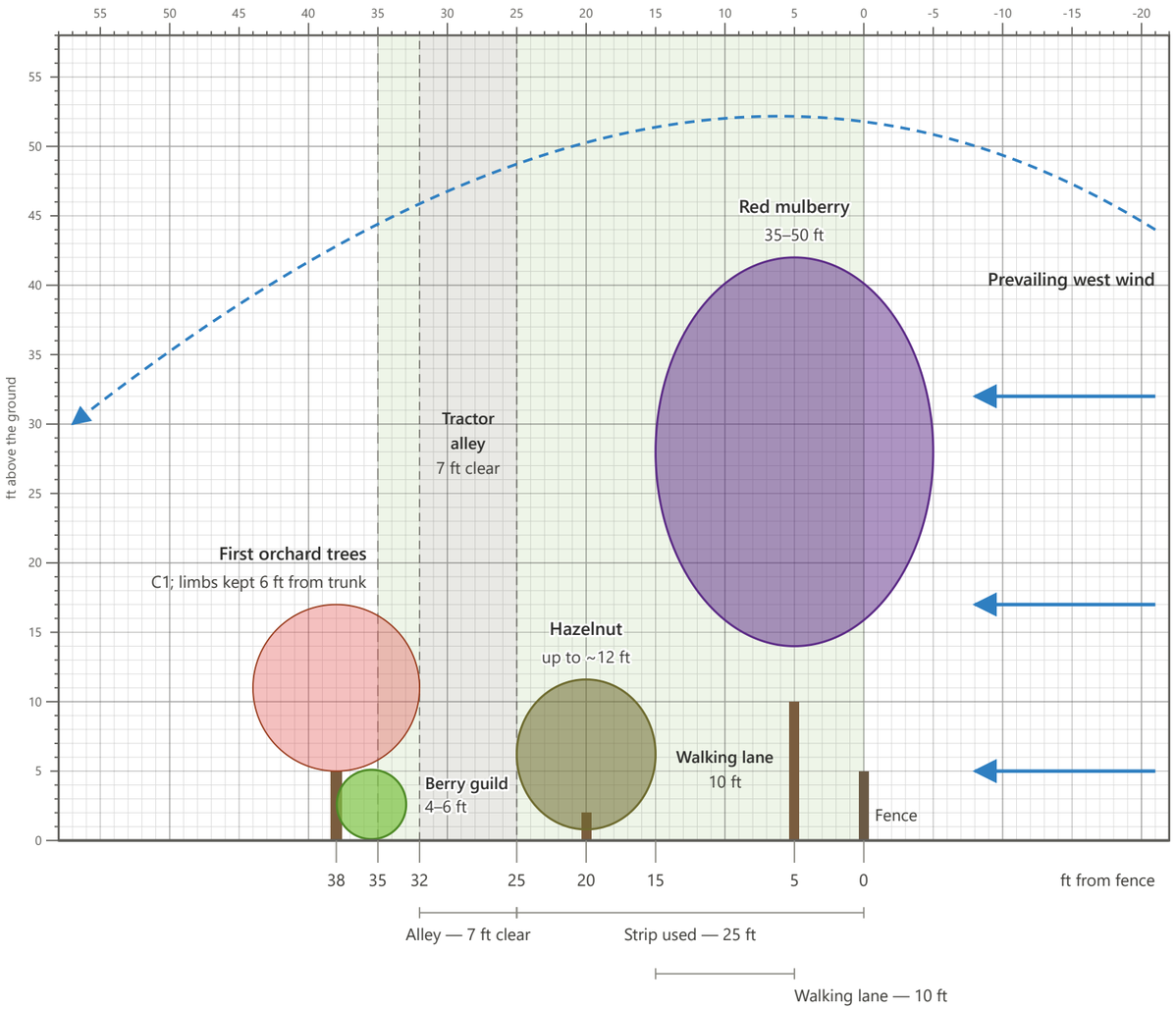


Red mulberry · Hazelnut · Small berries · First-column orchard tree and its limb room · A place for a plant still to add · Trunks: the large dot is a mulberry, the small dot a hazelnut

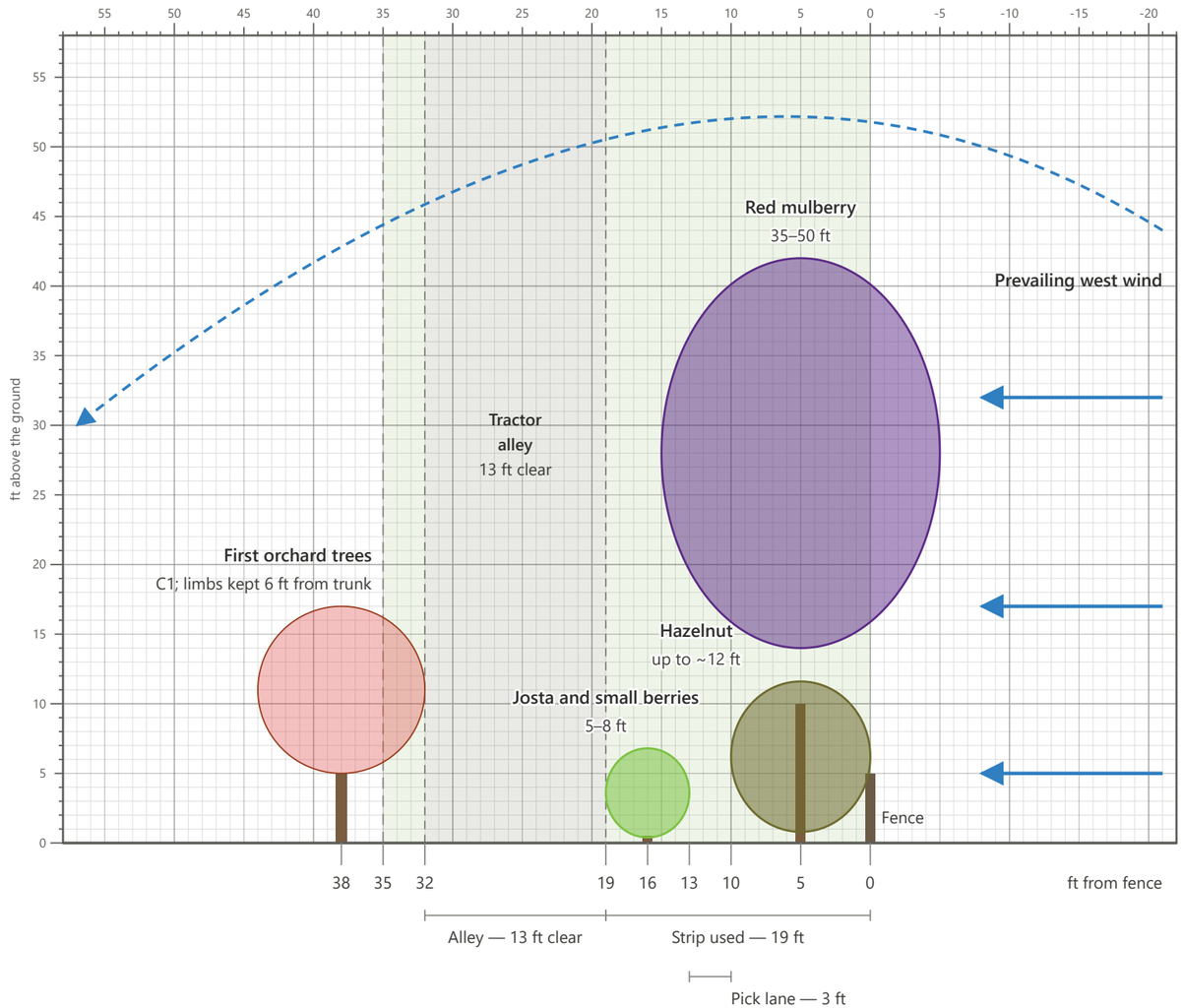
Stretch shown: 12 ft south to 48 ft north of the R1 line, beside the first three orchard rows. The top ruler reads feet from the fence; the left ruler reads feet north of the R1 line. The measurements west of the fence are the distance from one mulberry to the next, from one hazelnut to the next, and, in the proposed rows, from one berry bush to the next. In the proposed rows the mulberry line has a tick at every plant between its ends, so the gap from a mulberry to the hazelnut beside it reads off the same line. Every distance is in whole feet; where the current plan falls between two feet, the figure is rounded up. The berry bushes stand in runs by kind, honeyberries first: HB honeyberry, CU red currant, GB gooseberry, AR Viking aronia, JO jostaberry. Distances are to plant centers; the circles are the managed widths the plan already uses.

Profile — looking south, heights to scale

Now: mulberry, walking lane, hazelnut, alley



Proposed: pattern B (M–H–H), one row and a berry row



From the side the plants of a row overlap, so one crown of each kind stands for the row. The mulberry is drawn at the plan's own size: 35–50 ft tall, trunk to 10 ft, crown from 14 ft, kept to 20 ft wide. The top ruler reads feet from the fence; the left ruler reads feet above the ground.

The layers, west to east

Layer	Plant	Count	From west fence	In-row spacing	First plant	Last plant	Mature size	Notes
1. Tall layer in the guild row	**Red mulberry** Morus rubra	18 7 of the 25 on hand spare	5 ft	18 ft, every third place	13 ft south of the R1 line	13 ft north of the R15 line	35–50 ft tall; 35–40 ft wide naturally; keep about 20 ft	The main wind barrier, now in every third place along the row, 18 ft apart: wider than the 6–15 ft in-row range UNL gives for deciduous windbreak trees, but the hazelnuts and berries fill the space between, and the crowns still meet. Keep the plan's pruning and keep it up: one trunk, limbs from about 14 ft, and the east side cut back to about 10 ft from the trunk, so the crown hangs over only the nearest 2 ft of the berry row, 14 ft or more up (the light check on the options page shows why). Seedlings are usually male or female, so expect some fruitless trees (MOBOT).
2. Shrub layer in the same row	**Hazelnut** Corylus hybrids	34	5 ft	6 ft	7 ft south of the R1 line	7 ft north of the R15 line	Up to about 12 ft tall; nearly as wide as tall; keep about 10 ft	Two between each pair of mulberries, 6 ft apart, inside the 6–9 ft UMHDI gives for nut production. Each stands 6 ft from the next place in the row, so the hazelnut beside a mulberry is 6 ft from its trunk. They no longer stand 15 ft from the mulberries, the gap the USDA hybrid hazelnut guide keeps so the hazels are not shaded. The light check puts them at 70% of the season's sun with the mulberries kept as planned, against 74% in two rows, and 38% against 55% if the mulberries spread. Pick from the pick lane on their east side, and layer low stems into any gaps.
3. Berry row in front	**Small berries** honeyberry 11, red currant 7, gooseberry 3, Viking aronia 2, jostaberry 18	41 planted + 10 to add 41 on hand	16 ft	6 ft, between the trunks	10 ft south of the R1 line	230 ft north of the R1 line (last planted); open places to 10 ft north of the R15 line	5–8 ft tall; 5–8 ft; keep about 6 ft	One bush stands in front of every gap between two places in the row, 6 ft apart and 3 ft out of line with the trunks, so each bush is between two trunks. Each kind stands in a run, south to north, in the order the C1 guilds use (honeyberry, red currant, gooseberry, Viking aronia, jostaberry): the honeyberries need another cultivar to set fruit, so they stay one group, and one net covers a run. Josta is 5–8 ft tall and wide (cultivar library). These are the 41 bushes now planned around the C1 trees; 10 more places stay open at the north end. Honeyberry, currant, gooseberry, and aronia are smaller and could stand closer.
The east elderberry hedge is not part of this mock-up and stays as planned.								

Other small plants in your inventory

Honeyberry is already in the front row: all 11 are among the 41 bushes, standing together as one group. The table below adds the other small plants on your inventory sheet that the plan does not place yet, with where each would go. The front row is for bushes that are thornless or nearly so, do not run, and put up with part shade, next to a pick lane and a tractor alley. The rest do better in a bed or block of their own.

Plant	On hand	In the front row?	Where it would go	Why
Honeyberry bought: Blue Diamond ×3, Blue Palm ×3, Indigo Treat ×2, Cinderella ×2, Maxine, Sunrise	11	Yes, as one group	In the front row, as the first run at the south end, with the varieties mixed.	11 at Prairie, 1 fewer than purchased. The Berry sheet says they need another honeyberry, so plant them as one group and mix the varieties. The flowers open very early and tolerate frost to about 19–20°F, the bushes scorch in summer heat, so the row's afternoon shade suits them, and birds are the main pest, so one run takes one net.
Comfrey perennial, not a bush	25	No	A bed of its own, in ground you will not dig again. To have its mulch close to the berries, put the bed just past the north end of the row.	A clump that passes 3 ft, from a deep, thick root system that resprouts, so it is hard to remove. It does not belong beside the pick lane or the alley, where you may dig or reshape the ground. The sheet says yours is the sterile-seed kind, so it sows no seedlings and there is no need to cut the flower stalks; it spreads only from roots that are cut or dug up (UMN). Not for eating (cultivar library).
Rhubarb perennial, not a bush	15	No	A garden bed.	A 3 ft clump that stays 10 to 15 years or more and likes moist, well-drained soil. It would take front-row places a berry could use. Cut off the leaf blades: they are toxic.
Sea Berry berry	11	No	A block of its own in the open, well away from the alley and the orchard.	10 to 15 ft tall and 8 to 10 ft wide, armed with thorns, and it suckers into thickets: wrong beside a tractor alley and a pick lane. Only the females fruit, and the mix of sexes on hand is not recorded.
Blackberry various	3	No	A bed with a trellis wire, away from the alley.	The canes sprawl and sucker into what is beside them, thorns depend on the cultivar (not recorded), and the Idaho bulletin calls trailing types risky here; erect types are hardier.
Rasberry? berry	1	No	In the bramble bed, to see what it is.	One plant of an unknown kind. Brambles sucker into what is beside them.
Hardy Kiwi berry	4	No	A trellis or pergola near the house or the garden.	A twining vine that can add about 30 ft in a year and needs a sturdy trellis. Fruit needs a male and a female, and the sex and species of the 4 on hand are not recorded.
Mint perennial, not a bush	18	No	Where it is.	The sheet says: all planted. Mint runs, so keep it out of the strip.

How it is laid out

The mulberries stay 5 ft from the fence, where they are in the plan today. The hazelnuts stand in the same row, two hazelnuts between each pair of mulberries, with every place 6 ft from the next along the row, so each hazelnut is 6 ft from the trunk of the mulberry beside it and one guild is 18 ft of fence. A row of small berries stands 16 ft from the fence, with a 3 ft pick lane between it and the hazelnuts. Everything now ends 19 ft from the fence instead of 25.

What you gain

- **A wider alley.** The tractor alley is 13 ft clear instead of 7 (12 ft against 6 ft if every plant reaches its full width). About 1,750 sq ft is freed along the 292 ft of fence, and all of it goes to the alley: the orchard trees are in the ground, so the orchard does not move.
- **No dead lane.** The 10 ft walking lane, and the 15 ft shade gap that called for it, become a 3 ft pick lane.
- **One barrier, dense from the ground up.** The hazelnuts fill the open space under the mulberry crowns, so the row blocks wind near the ground as well as above it. Red mulberry grown in the open tends to have low limbs, and the hazelnuts cover the base while you lift them.
- **A home for the berries.** All 41 berry bushes planned around the C1 trees move to the lee of the row, and the C1 trees no longer need guilds around them. Each kind stands in a run, so the honeyberries are one group and a net covers a run. Only the 3 gooseberries are likely to have thorns (their cultivar is not recorded); the rest are thornless, so the row is easy to pick.

What it costs

- **Hazelnut light depends on the mulberry crowns.** With the mulberries kept as the plan assumes (20 ft wide, limbs from 14 ft), the hazelnuts get 74% of the season's direct sun now and 70% in pattern B (M–H–H). If the mulberries are left to spread (30 ft wide, limbs from 8 ft), it is 55% now and 38% here. The guild row works only with firm pruning: one trunk, limbs from 14 ft, and the east side cut back to 10 ft from the trunk. How the light is worked out.
- **Fewer mulberries, or fewer hazelnuts.** This pattern plants 18 mulberries (25 now; 7 of the 25 on hand spare) and 34 hazelnuts (34 now; all 34 on hand used).
- **Tighter work space.** With no walking lane, mulberries are pruned and hazelnuts picked from the 3 ft pick lane or from the alley side. That takes a person and a cart, not a mower.
- **Shade and snow on the berries.** The row throws afternoon shade on them: about 73% of the season's sun reaches the berry row. West winds drop their drifts just east of the row, over the berries and the alley, as the water plan already expects for the hazelnut row.

What else it touches

- **The water plan.** Zone 9 makes the walking lane a sunken wood-chip path between the two rows; here that job moves to the pick lane. Zone 8's water bars turn water west into the hazelnut row, which becomes the guild row, and zone 7's drifts land on the berry row. See the water-harvesting plan.
- **The income page.** It counts 25 mulberries and 34 hazelnuts; see crop and producer income.
- **The wind.** No wind has been measured at the lot, and the stations disagree about winter wind direction. A west windbreak, in either layout, is a bet that the winter wind blows from the west; the site baselines say to wait for a year of weather-station readings.

Decisions for you

- **Which pattern.** A keeps the most mulberries but gives the hazelnuts the least light. C gives the hazelnuts the most light and uses the fewest mulberries. B sits between. Choose above and the figures redraw.
- **Where the berries go.** The figures move all of them to the front row. Splitting them, some here and some around the C1 trees, leaves places open in the row; keep the honeyberries together wherever they go. Comfrey, rhubarb, and the brambles are left out of the row.
- **How firmly you will prune.** The numbers above assume the plan's pruning. A mulberry left to spread costs the hazelnuts far more in the guild row than in two rows.

The current plan is read from `data/orchard.json` through the orchard model (`data/orchard_model.js`), so the left-hand figures follow the data. The proposal's own choices, a 6 ft step along the row, a 3 ft pick lane, and a 6 ft berry row, are in `windbreak_guild_model.js`. Josta grows 5–8 ft tall and wide (cultivar library). Widths are the managed widths the plan already uses: mulberry 20 ft, hazelnut 10 ft.